

AI MARKETING INTELLIGENCE SYSTEM FOR AMAZON PRODUCT REVIEWS

This project presents an AI-based marketing decision system built with Python and Power BI.

The system analyzes Amazon product reviews to understand customer satisfaction, segment customers, predict positive reviews, recommend products, and support better marketing decisions.

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BUSINESS PROBLEM AND PROJECT GOAL

Modern companies receive thousands of customer reviews, but it is difficult to analyze them manually.

Marketing teams need to understand what customers like, what problems they mention, and which products should be promoted.

The goal of this project is to build an AI marketing system that helps managers make faster and better decisions.

Our system helps to:

- understand customer behavior
- find customer segments
- predict positive customer feedback
- recommend products by segment
- analyze customer sentiment.

DATASET AND TOOLS

For this project, we used an Amazon product reviews dataset.

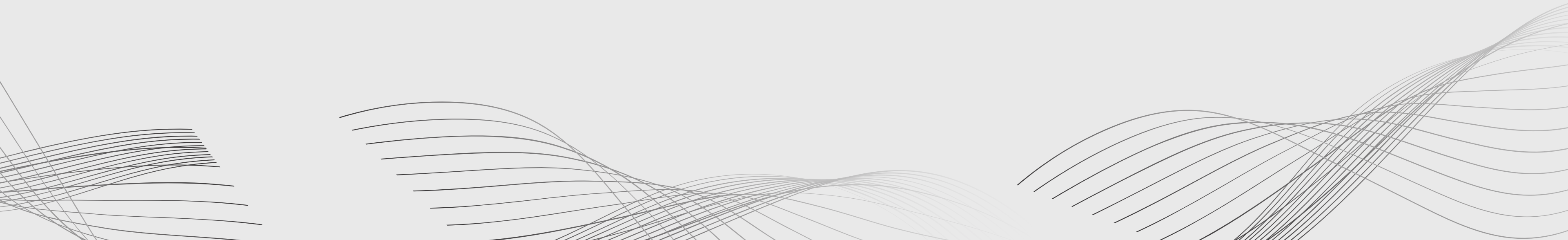
The dataset includes customer reviews, product IDs, user IDs, review scores, review text, helpfulness information, and review dates.

Main columns used in the project:

- ProductId
- UserId
- Score
- Summary
- Text
- Time
- HelpfulnessNumerator
- HelpfulnessDenominator

Tools and libraries used:

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn
- TextBlob
- WordCloud
- Power BI



PYTHON NOTEBOOK: DATA CLEANING AND EDA

In the Python notebook, we first cleaned and prepared the dataset.

We removed duplicates, handled missing values, converted time into a normal date format, and created new useful columns.

New features included:

- ReviewYear
- ReviewMonth
- ReviewLength
- HelpfulnessRatio
- ScoreSentiment
- PositiveReview

After cleaning, we performed exploratory data analysis.

We analyzed review score distribution, sentiment distribution, review activity over time, top products, active customers, and correlation between numeric variables.

The main finding was that most reviews were positive, which means customer satisfaction was generally strong.



CUSTOMER SEGMENTATION AND PREDICTION MODEL

We used K-Means clustering to divide customers into meaningful segments based on their review behavior.

The main customer segments were:

- Loyal Positive Reviewers
- Low Engagement Customers
- Critical Customers

This helps the marketing team create different strategies for different customers instead of using one general campaign for everyone.

We also built a predictive model using Random Forest.
The model predicts whether a review will be positive or not.

The model used features such as:

- review length;
- helpfulness ratio;
- review year and month;
- customer activity;
- customer segment.

This helps the company understand which factors are connected to positive customer experience.

RECOMMENDATION SYSTEM AND SENTIMENT ANALYSIS

We created a simple recommendation system based on product performance and customer segments.

The system recommends products with higher review scores and better helpfulness results.

This can help the company personalize marketing offers for different customer groups.

We also performed sentiment analysis using customer review text.

Reviews were classified as:

- Positive
- Neutral
- Negative

The sentiment analysis helps the marketing team understand customer emotions, identify common complaints, and use positive customer language in marketing campaigns.



POWER BI DASHBOARD OVERVIEW

The Power BI dashboard contains 8 pages:

- 1.Executive Marketing Overview
- 2.Customer Segmentation Dashboard
- 3.Predictive Analytics Dashboard
- 4.Recommendation Dashboard
- 5.Product Performance Dashboard
- 6.Sentiment Analysis Dashboard
- 7.Ethics, Privacy & Risk Dashboard
- 8.AI Marketing Strategy

The dashboard includes KPI cards, bar charts, line charts, donut charts, tables, slicers, and interactive filters.

The dashboard allows marketing managers to quickly understand customer satisfaction, product performance, customer segments, sentiment trends, and final business recommendations.

FINAL RECOMMENDATIONS AND BUSINESS VALUE

Based on our analysis, we suggest five main marketing recommendations.

First, the company should focus on loyal positive reviewers because they create strong positive feedback and support brand trust.

Second, products with negative reviews should be improved by analyzing complaints from critical customers.

Third, personalized recommendations should be used for different customer segments.

Fourth, products with high scores and many reviews should be promoted more actively.

Fifth, sentiment trends should be monitored regularly to detect customer dissatisfaction early.

Overall, this AI marketing system helps the company understand customers better, improve product promotion, increase engagement, and make more data-driven marketing decisions.

THANK YOU

